The Dark-Matter Effect as Large-Scale Field Organization within the Framework of the Universal Quantum Foam Hypothesis (UQSH) V2 12.03.26

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Abstract

The dynamics of galaxies and galaxy clusters exhibit gravitational effects that cannot be explained solely by observable baryonic matter. In the standard model of cosmology, these observations are described by an additional matter component referred to as dark matter.

This work proposes an alternative interpretation within the framework of the Universal Quantum Foam Hypothesis (UQSH). It is assumed that physical space can be described by a universal continuous field medium with the local organizational state $\phi(x, t)$. Within this framework, matter appears as bound field organization, while radiation creates real field excitations and deformations of the medium.

The observed dark matter effect is interpreted as a consequence of large-scale, persistent field organization arising from

the long-term superposition of materially bound and radiation-induced deformation states. Additional gravitational effects thus do not necessarily arise from additional particles, but as emergent properties of the field structure.

The work formulates a conceptual and minimal field-mechanical framework for this interpretation and discusses phenomenological consequences for galactic rotation curves and cosmological structures.

Contents

1 Introduction

Since the early observations by Zwicky and subsequent investigations of galaxy rotation curves [1, 2, 3], it has been known that gravitational effects occur on cosmic scales that cannot be explained solely by visible matter.

In particular, spiral galaxies exhibit rotation curves in which the orbital velocity of stars remains nearly constant even at large radial distances. According to the expectations of classical gravitation, the velocity should instead decrease with increasing radius.

To explain this discrepancy, the standard model assumes an additional matter component that has no electromagnetic interaction and is therefore not directly observable.

Despite intensive experimental searches, no unambiguous particle candidate for dark matter has been identified to date. This situation has led to the development of alternative theoretical approaches.

The present work pursues such an approach within the framework of the Universal Quantum Foam Hypothesis (UQSH), in which gravitational effects are interpreted as structural properties of an underlying field medium.

This work does not claim to formulate a complete cosmological theory. Rather, its goal is to interpret the observed dark matter effect within the UQSH framework as a consequence of large-scale field organization and to provide a minimal field-mechanical mechanism for its emergence.

2 Observational Evidence

Several independent astrophysical observations point to additional gravitational effects.

2.1 Empirical Evidence from Rotation Curves

Measurements of the rotation velocity of stars show that it remains nearly constant at large radial distances from the galactic center.

This suggests that the effective gravitational mass increases with growing radius.

2.2 Galactic Rotation Curves

For the phenomenological description of these observations, we use an effective decomposition of the rotational dynamics.

The orbital velocity of a star at distance r from the galactic center is given by

$$v(r)^2 = r \, \frac{d\Phi_{\text{eff}}}{dr}. \tag{1}$$

with

$$\Phi_{\text{eff}}(r) = \Phi_{\text{bar}}(r) + \Phi_{\text{LS}}(r), \tag{2}$$

where Φ_{LS} describes the contribution of large-scale field organization.

This yields phenomenologically

$$v(r)^2 = v_{\text{bar}}^2(r) + v_{\text{LS}}^2(r). \tag{3}$$

This formally corresponds to the decomposition of rotation curves into baryonic and additional components frequently used in observations. Within the UQSH, however, the additional contribution is not interpreted as a new matter component, but as an effect of large-scale field organization.

2.3 Gravitational Lensing

Gravitational lensing measurements also show that galaxies and especially galaxy clusters possess a significantly larger gravitational effect than can be derived from visible matter alone.

2.4 Cosmological Structure Formation

Numerical simulations of large-scale structure formation in the universe successfully reproduce observed structures only when additional gravitational effects are taken into account.

3 Fundamental Assumptions of the Universal Quantum Foam Hypothesis

The Universal Quantum Foam Hypothesis is based on the assumption that physical space can be described by a universal continuous field medium.

Within this framework, the following fundamental assumptions apply:

- Physical space possesses a continuous field-like structure.
- Matter can be interpreted as a stable organizational form of this field medium.
- Gravitational effects arise from structural gradients or tension states within the field.

These assumptions enable an alternative interpretation of gravitational phenomena. In particular, they open the possibility of interpreting large-scale additional gravitational effects not as an expression of additional matter, but as a consequence of persistent organizational states of the field medium.

4 Emergence of the Dark Matter Effect in the Quantum Foam

Within the framework of the Universal Quantum Foam Hypothesis, the dark matter effect does not arise from an additional, invisible matter substance, but through the long-term field-mechanical organization of the quantum foam itself.

In the context of the present work, the term quantum foam refers to the continuous, dynamic, and deformable field medium that constitutes physical space.

The starting point is the assumption that physical space is a continuous, elastic, and reactive field medium. This medium is not empty, but is in a permanently excited state due to matter, motion, and especially radiation. Within the UQSH, radiation is not understood as mere energy transport through empty space, but as a real excitation of the field that produces local deformations, tensions, and restructurings.

Each baryonic structure therefore acts not only through its local mass distribution, but simultaneously through its effect on the tension state of the field. Stars, gas clouds, galactic centers, jets, and other radiation-active processes shape the quantum foam not only locally, but on large scales. Where many such influences act together over long time periods, superimposed deformation and tension profiles of the medium emerge.

This superposition is not simply additive, but geometric. Field deformations can amplify, redirect, partially dissipate, or transition into new patterns. This does not create sharply bounded "objects," but extended zones of common organization. Galactic halos appear in this picture not as accumulations of invisible particles, but as large-scale superposition spaces of persistent field deformations.

The triggering mechanism of the dark matter effect is thus the field-mechanical prestressing of space through sustained excitation, coupled with the long-term storage and superposition of these deformations in the quantum foam. The observed additional gravitational effect is the dynamic response of the field to this translocal tension state.

Baryonic matter remains central, but not as the sole source of gravitation in the classical sense, but as a coherent anchor of field organization. Therefore, the dark matter effect follows the visible structure closely in many systems, without being identical to it. The additional gravitation arises from the large-scale response of the field to stable material and radiation-induced patterns.

In this sense, the so-called dark matter halo is not a substance and not a particle reservoir, but a metastable organizational state of the quantum foam, fed by past and present field excitation.

5 Field-Mechanical Origin of the Dark Matter Effect

Within the framework of the Universal Quantum Foam Hypothesis, physical space is described as a continuous, dynamic field medium. The local organizational state of this medium is characterized by a scalar field

$$\phi(x,t) \tag{4}$$

This field describes the local structure, tension, and organization of the underlying spatial field.

5.1 Dynamics of the Field

The temporal evolution of field organization can be described in minimal form by a nonlinear wave equation

$$\partial_t^2\phi - c^2\nabla^2\phi + V'(\phi) = J_{\text{bind}}[\phi] - J_{\text{relax}}[\phi] + J_{\text{free}}[\phi]. \tag{5}$$

The individual terms describe different physical processes:

- $J_{\text{bind}}[\phi]$ describes bound field organization through stable material structures.
- $J_{\text{relax}}[\phi]$ describes local relaxation processes of the field.
- $J_{\text{free}}[\phi]$ describes free field excitations, particularly through radiation.

The equation used here is to be understood as minimal field-mechanical modeling and serves the conceptual description of the organizational, relaxation, and excitation processes assumed in the UQSH.

It is to be read as an extended effective regime decomposition and does not replace the minimal basic form of field dynamics, but rather concretizes its possible binding, relaxing, and free organizational components.

For large-scale galactic structures, a quasi-stationary regime can often be assumed, in which rapid temporal changes in field organization are negligible compared to their spatial structure. In this limit case, the full dynamic equation is replaced by an effective stationary description, in which large-scale field organization can be characterized by an elliptic equation.

5.2 Matter as Bound Field Organization

Within this framework, matter does not appear as an independent ontological category, but as a stable organizational form of the field. An effective bound organizational density can be described by

$$\rho_{\text{bind}}(x) = \kappa_b\left(\frac{1}{2}\phi(x)^2 + \frac{\ell^2}{2}|\nabla\phi(x)|^2\right) \tag{6}$$

This quantity characterizes locally stable organizational states of the field medium.

The chosen parametrization is to be understood as a simplified effective representation of bound field organization and serves here for phenomenological modeling.

In the stationary regime, the field-shaping effect of bound structures can be set proportional to this bound organizational density as a first approximation. Thus, $\rho_{\text{bind}}(x)$ forms a natural effective source of large-scale field organization.

5.3 Radiation-Induced Field Excitation

Within the UQSH, radiation does not represent a movement of energy through empty space, but a real excitation of the underlying field. Radiative processes therefore create local deformations and tension states in the field medium.

The local field state can therefore be heuristically described as a superposition of several contributions

$$\phi(x,t) = \phi_{\text{bind}}(x,t) + \phi_{\text{rad}}(x,t) + \phi_{\text{LS}}(x,t). \tag{7}$$

Where

- ϕ_{bind} denotes bound organizational structures through matter,
- ϕ_{rad} radiatively generated field excitations,
- ϕ_{LS} large-scale persistent field organization.

5.4 Gravitation as Gradient of Field Organization

Gravitational dynamics arises from gradients of the field state. The effective gravitational potential can be defined by

$$\Phi(x) = \alpha\phi(x) \tag{8}$$

The resulting acceleration is given by

$$\vec{a}(x) = -\nabla\Phi(x) = -\alpha\nabla\phi(x). \tag{9}$$

Gravitation thus appears as a dynamic response to spatial gradients of the organizational state of the field.

5.5 Large-Scale Field Organization and Dark Matter

Through long-term interactions between baryonic matter, dynamic processes in galaxies, and continuous radiation excitation, large-scale field structures can emerge. This persistent field organization is described by the term ϕ_{LS}.

The observed dark matter effect can be interpreted within this framework as an effective gravitational effect of this large-scale field organization.

An effective density of the dark matter contribution can be formally written as

$$\rho_{\mathrm{DM}}^{\mathrm{eff}}(x) = \eta_{\mathrm{DM}}\, B[\phi](x) \tag{10}$$

where $B[\phi]$ describes a functional that captures the large-scale superposition, persistence, and geometric organization of field deformations.

An explicit mathematical form of this functional is not yet specified in the present work. $B[\phi]$ serves here as an open formal representation of that large-scale field structure that is held responsible for the observed additional gravitational effect.

In this interpretation, the so-called dark matter halo does not represent an additional matter component, but a metastable large-scale organizational state of the field medium.

Phenomenologically, the resulting large-scale contribution to gravitational dynamics can be described in the following sections by an effective additional potential Φ_{LS}.

6 Phenomenological Modeling of Field Organization

To phenomenologically describe the additional gravitational effect resulting from large-scale field organization, an effective additional potential can be introduced.

We write the total potential as

$$\Phi_{\mathrm{eff}}(r) = \Phi_{\mathrm{bar}}(r) + \Phi_{\mathrm{LS}}(r), \tag{11}$$

where Φ_{bar} describes the potential of baryonic matter and Φ_{LS} describes the contribution of large-scale field organization.

This term does not represent an additional matter component, but a phenomenological representation of the large-scale field organization described in the previous section.

A simple phenomenological form can be approximated by

$$\Phi_{\mathrm{LS}}(r) \propto v_0^2 \ln(r/r_0) \tag{12}$$

This form leads to a nearly constant rotation velocity

$$v(r) \approx v_0 \tag{13}$$

for large radii and thus qualitatively reproduces the observed flat rotation curves of many spiral galaxies.

Within this framework, the additional gravitational effect can be interpreted as a consequence of a large-scale organizational state of the underlying field medium.

6.1 Asymptotic Relaxation Regime of the Field

Within the UQSH framework, it is assumed that baryonic matter and radiation processes directly shape the local organizational state of the field in the inner region of a galactic system. With increasing distance from the baryonic core, the local field-shaping effect of matter and radiation decreases. The field itself does not vanish, but transitions into a relaxation regime determined by the boundary conditions of large-scale field organization.

For an effective stationary description we write

$$\nabla^2 \phi \approx -S_{\mathrm{eff}}(x), \tag{14}$$

where $S_{\mathrm{eff}}(x)$ summarizes the local field-shaping effect of baryonic structure and radiation excitation.

The term $S_{\mathrm{eff}}(x)$ denotes an effective source structure in which the local field-shaping effect of bound matter organization, radiative excitation, and relaxing counter-processes are combined.

Conceptually, S_{eff} can thus be understood as a stationary regime formulation of the previously introduced dynamic contributions J_{bind}, J_{relax}, and J_{free}.

In the large-scale outer region of a galactic system, this source term becomes small, so that asymptotically

$$S_{\text{eff}}(x) \to 0 \qquad \text{and thus} \qquad \nabla^2\phi \approx 0 \tag{15}$$

holds.

The large-scale halo regime appears in this view as a relaxation region of the field, in which no longer the immediate local matter shaping, but the free spatial organization of the field state dominates.

6.2 Logarithmic Field Profile and Flat Rotation Curves

For an effective rotationally symmetric description of the large-scale outer region of a galactic system, a logarithmic profile represents a simple asymptotic model solution:

$$\phi(r) = A \ln\left(\frac{r}{r_0}\right), \tag{16}$$

where A is a constant and r_0 a reference length.

With the UQSH coupling

$$\Phi(r) = \alpha\phi(r) \tag{17}$$

it follows that

$$\Phi(r) = \alpha A \ln\left(\frac{r}{r_0}\right). \tag{18}$$

The rotation velocity on a circular orbit is given by

$$v(r)^2 = r\,\frac{d\Phi}{dr}. \tag{19}$$

Since

$$\frac{d\Phi}{dr} = \frac{\alpha A}{r}, \tag{20}$$

one immediately obtains

$$v(r)^2 = \alpha A. \tag{21}$$

Thus it follows

$$v(r) = \sqrt{\alpha A} = \text{const.} \tag{22}$$

An asymptotically logarithmic profile of large-scale field organization thus directly leads to flat rotation curves.

6.3 Connection to the Baryonic Tully–Fisher Relation

One of the most striking empirical relationships in galaxy physics is the so-called baryonic Tully–Fisher relation. Observations show that the asymptotic rotation velocity of spiral galaxies is closely linked to their total baryonic mass. Empirically, approximately

$$M_b \propto v^4, \tag{23}$$

holds, where M_b denotes the total baryonic mass of a galaxy and v the asymptotic rotation velocity in the outer region of the rotation curve.

Within the UQSH framework, this relation can be interpreted as a consequence of the field-mechanical coupling between baryonic structure and large-scale field organization.

In the outer halo region of a galactic system, the field state transitions into a relaxation regime in which the effective source shaping by baryonic matter and radiation has strongly decreased. In this region, the field can be approximately described by a source-free equation

$$\nabla^2 \phi \approx 0. \tag{24}$$

A possible asymptotic solution of this equation is a logarithmic profile of the form

$$\phi(r) \sim A \ln r, \tag{25}$$

where A describes the large-scale amplitude of field organization.

Using the relationship between gravitational potential and field state introduced in this work

$$\Phi = \alpha \phi, \tag{26}$$

the rotation velocity is given by

$$v^2 = r \frac{d\Phi}{dr} = \alpha A. \tag{27}$$

Thus, the asymptotic rotation velocity is directly determined by the amplitude of large-scale field organization.

Within the UQSH, it is assumed that this amplitude is determined by the global baryonic structure of the system. A dimensionally motivated heuristic assumption is that the field amplitude scales with the square root of the baryonic mass,

$$A \propto \sqrt{M_b}. \tag{28}$$

This immediately yields

$$v^4 \propto M_b. \tag{29}$$

The baryonic Tully–Fisher relation thus appears in this picture not as a coincidental correlation of two independent components, but as a natural consequence of the coupling between baryonic field organization and large-scale relaxation state of the underlying field medium [7, 8].

6.4 Saturation Limit of Field Organization

Within the UQSH framework, it is assumed that the underlying field medium is deformable but not arbitrarily compressible. Local tension states of the field can therefore only grow up to a certain maximum organization.

Formally, this can be interpreted as a limitation of local field gradients,

$$|\nabla\phi| \leq S_{\max}, \tag{30}$$

where $S_{\max}$ describes a characteristic saturation limit of field organization. This quantity is understood here as an effective material or structural property of the underlying field medium.

When this limit is reached, additional field-shaping effects can no longer be absorbed by further local increase of field tension. Instead, the field reorganizes spatially, so that the field structure extends over larger scales.

In astrophysical systems, this means that increasing baryonic mass does not create unboundedly steeper local field gradients. Rather, the spatial extent of field organization grows, whereby extended halo structures form.

This saturation dynamics offers a natural mechanism for limiting central field tensions and can thus also contribute to explaining flat central density profiles in many galaxies.

6.5 Characteristic Acceleration Scale

The saturation limit of field organization introduced in the previous section implies a natural restriction of locally achievable field gradients.

Since within the UQSH framework gravitational acceleration is determined by the gradient of the field state,

$$\vec{a} = -\alpha \nabla \phi, \tag{31}$$

a maximum field tension immediately leads to a characteristic acceleration scale

$$a_0 = \alpha S_{\max}, \tag{32}$$

where $S_{\max}$ describes the maximum local field organization.

This scale marks the transition between two dynamical regimes. For accelerations

$$a \gg a_0 \tag{33}$$

the immediate baryonic field shaping dominates, while for

$$a a_0 \tag{34}$$

the large-scale relaxation of the field becomes increasingly determining.

The observed dark matter effect can thus be interpreted in this picture as a transition between a locally shaped and a relaxed field state.

6.6 Limitation of Halo Structure on Large Scales

The logarithmic field profile

$$\phi(r) \sim A \ln r \tag{35}$$

represents only an asymptotic description of the relaxation regime. Formally, the logarithm grows without upper limit. In real astrophysical systems, however, such a divergence does not occur, since galactic halos do not represent isolated systems.

On large scales, the field organization of a galaxy is limited by several physical effects:

- Superposition with the field structures of neighboring galaxies,
- Embedding in the large-scale structure of galaxy groups and galaxy clusters,
- The cosmological background of the field medium.

In the language of the present field description, this means that the logarithmic relaxation profile is only valid in a limited radial range. Beyond a characteristic scale R_{halo}, the galactic field profile continuously transitions into the superordinate field organization of the cosmic environment.

This can be formally interpreted as a transition of boundary conditions. While in the inner region of a galactic system the baryonic structure shapes the field organization, on large scales the cosmic field environment increasingly becomes dominant.

The logarithmic profile therefore does not describe an unboundedly growing potential, but a transition regime between local baryonic field shaping and large-scale cosmic field organization.

A complete quantitative determination of R_{halo} would, however, require explicit modeling of the coupling between galactic and cosmic field organization and remains reserved for further work.

6.7 Compatibility with Gravitational Lensing

A central test of any interpretation of the dark matter effect is its compatibility with gravitational lensing. Within the UQSH framework, the lensing effect does not follow the distribution of an additional invisible matter substance, but the effective geometry of the underlying field medium.

Since the effective gravitational potential is given by

$$\Phi(x) = \alpha\phi(x) \tag{36}$$

the light deflection is determined by the large-scale field organization. Formally, the deflection angle can be described as the transverse effect of the potential integrated along the line of sight,

$$\Delta\theta \propto \int \nabla_{\perp}\Phi \, dl, \tag{37}$$

where $\nabla_{\perp}\Phi$ denotes the transverse gradient of the effective potential. With $\Phi = \alpha\phi$ it follows that

$$\Delta\theta \propto \alpha \int \nabla_\perp \phi \, dl. \tag{38}$$

The lensing effect appears in this view as an integrated effect of transverse field organization along the line of sight. Thereby the observed light deflection can systematically be stronger than would be expected from visible baryonic matter alone, without an additional particle component being presupposed.

Particularly on the scales of galaxy clusters, where many gravitationally effective structures superimpose, this integrated field effect provides a natural framework for enhanced gravitational lensing. The UQSH thereby remains compatible with the macroscopic structure of General Relativity, but interprets the effective geometry as an expression of large-scale field-mechanical organization.

7 Central Observational Reference Points

In the following, some particularly important astrophysical findings are mentioned at which the field-mechanical interpretation of the dark matter effect within the UQSH framework can be concretely illustrated. The goal of this section is not to provide a complete phenomenological derivation, but to show that several central observations can be read within a common field-mechanical interpretive framework.

7.1 Radial Acceleration Relation and Baryonic Coupling

The so-called Radial Acceleration Relation (RAR) describes the close relationship between the acceleration expected from the baryonic matter distribution and the actually observed dynamical acceleration in galaxies [7]. The additional acceleration, which in the standard picture is attributed to a dark matter component, follows the baryonic structure in a strikingly close manner.

Within the ΛCDM framework, this coupling is usually understood as a result of complex galaxy formation and feedback processes. Within the UQSH framework, however, the same observation can be interpreted as a direct consequence of a common field-mechanical basis. If baryonic structures themselves represent stable organizational patterns of the field medium, then they simultaneously shape the local and large-scale tension distribution of the field.

The close coupling between baryonic distribution and dynamical acceleration appears in this view not as subsequent fine-tuning, but as an expected consequence of the same underlying field organization. Compact baryonic structures should then create steeper tension profiles, while diffuse or orderly distributed structures lead to flatter additional effects.

7.2 Baryonic Tully–Fisher Relation

The baryonic Tully–Fisher relation connects the total baryonic mass of a galaxy with its asymptotic rotation velocity [8]. The observed precision of this relation represents an important structural feature within purely mass-based interpretations.

Within the UQSH framework, this relation can be understood as an expression of the field-mechanical coupling between stable baryonic organization and large-scale tension superposition. Larger baryonic masses correspond in this picture not only to a larger local amount of matter, but to a stronger and more extended field-mechanical organization. This deepens the large-scale tension gradient of the field that co-determines the observed rotational dynamics.

The baryonic Tully–Fisher relation thus appears as a natural correlate of a common field-mechanical basis of baryonic structure and additional gravitational effect. In this interpretation, it is not a coincidental correlation of two independent components, but an indication of their common structural origin.

7.3 Gravitational Lensing on Galactic and Cluster Scales

Gravitational lensing measurements show in many galaxies and galaxy clusters a stronger light deflection than can be derived from visible baryonic matter alone [10]. In the standard picture, this is considered particularly strong evidence for an additional gravitational source in the form of extended dark matter halos.

Within the UQSH framework, this finding can alternatively be read as an effect of integrated field geometry along the line of sight. In this model, light does not follow the distribution of an invisible substance, but the effective geodesics of a physically tensioned field medium. Where large-scale tension profiles superimpose, the effective field geometry deepens, so that stronger lensing effects can occur without an additional particle component being presupposed.

This interpretation is particularly relevant on cluster scales, where many gravitationally effective structures superimpose. The observed lensing effect can then be understood as an expression of cumulative field organization. A complete quantitative test of this interpretation would, however, require explicit modeling of large-scale field geometry along realistic lines of sight.

7.4 The Bullet Cluster as a Test Case for Differently Persistent Field Organization

The Bullet Cluster represents a particularly important test case for any non-standard interpretation of the dark matter effect [9]. What is observed there is a spatial separation between the hot baryonic gas, which remains in the collision region, and the centers of gravitational lensing effect, which lie closer to the galaxy populations.

In the standard picture, this is often interpreted as evidence for collisionless dark matter. Within the UQSH framework, the same finding can be read as an expression of differently persistent field organization. Hot gas also represents a field-mechanical state, but a diffuse and collision-prone one. In a dynamic collision, such a state loses part of its coherent large-scale field organization.

Compact baryonic structures like stars and galaxies, on the other hand, remain comparatively stable, bound organizational patterns of the field. Such stable patterns can carry the associated field-mechanical tension state longer and thus act as persistent anchors of large-scale field geometry.

In this interpretation, the center of gravitational effect does not necessarily follow the largest momentary baryonic gas mass, but the more stable and longer-lived field patterns. The Bullet Cluster would thus not necessarily be a counterexample to the UQSH, but a demanding test case for the question of how quickly large-scale field tensions reorganize and relax in highly dynamic collision situations. Whether this interpretation is quantitatively sufficient remains an open question for further modeling.

Precisely at this point it becomes clear that the UQSH currently provides primarily an interpretive and mechanistic framework, while quantitative field modeling of colliding cluster systems is still pending.

7.5 Central Profile Forms and the Core–Cusp Problem

Numerical simulations of cold dark matter frequently produce strongly rising density profiles in the centers of halos, while observations of many galaxies rather suggest flatter central cores [11, 12]. This discrepancy is known as the core-cusp problem.

Within the UQSH framework, a natural motivation for limited central profiles arises from the finite local elasticity or saturation capacity of the field medium. If additional gravitational effects are understood not as a distribution of additional particles, but as tension gradients of the field, an unboundedly growing central tension state is not to be expected. Above a critical local load, additional structure should not lead to arbitrarily further deepening, but to redistribution, extension, or saturation of the tension profile.

Flatter central cores appear in this view not as subsequently smoothed exceptions, but as field-mechanically plausible limit forms of stable organization. The UQSH thus provides at least qualitatively a natural interpretive framework for the occurrence of cored instead of cuspy central profiles.

7.6 Possible Time-Delayed Adjustment of Large-Scale Field Organization

If additional gravitational effects are interpreted as large-scale organizational states of a physical field medium, then changes in this field structure cannot occur instantaneously.

Since local field reorganizations are limited by the maximum reorganization rate c, a large-scale field organization could adjust to changed baryonic structures with a time delay.

In dynamic systems such as galaxy collisions or cluster interactions, additional gravitational effects could therefore show a certain inertia or post-organization. Such effects could in principle be observable and would represent a possible empirical distinction between particle-based dark matter and field-mechanical interpretations.

8 Comparison with Existing Interpretations

The gravitational anomalies on galactic scales are interpreted in different ways in the literature.

8.1 ΛCDM

In the standard model of cosmology (ΛCDM), it is assumed that a significant fraction of matter in the universe consists of non-baryonic dark matter. This matter interacts primarily gravitationally and forms large-scale halos around galaxies [6, 13].

This model reproduces many observations of cosmological structure formation, but presupposes the existence of particles not yet directly detected.

8.2 Modified Dynamics (MOND)

An alternative interpretation is provided by Modified Newtonian Dynamics (MOND). In this approach, it is assumed that the law of gravitation changes at very small accelerations.

MOND can successfully describe many rotation curves of individual galaxies, but encounters difficulties in describing cosmological structures and galaxy clusters.

8.3 Field-Based Interpretation within the UQSH Framework

The approach proposed in this work differs from both interpretations.

Instead of introducing additional particles or modified gravitational laws, it is assumed that the observed effects arise from large-scale organizational states of an underlying field medium.

In this picture, the dark matter effect can be interpreted as an emergent property of field structure.

9 Relationship to General Relativity

The Universal Quantum Foam Hypothesis does not understand itself as a replacement for General Relativity, but as a possible ontological interpretation of its geometric structure.

The Einstein equations successfully describe the relationship between energy-momentum distribution and spacetime curvature. Within the UQSH framework, this geometric description is interpreted as a macroscopic approximation of a deeper field-mechanical organization.

The local organizational state of the field $\phi(x)$ determines the effective field geometry in this view. Gravitational effects thereby appear as a consequence of spatial gradients of field organization, while the known relativistic phenomena of orbital curvature, time dilation, and gravitational lensing are preserved as macroscopic manifestations of this field structure.

Thus the empirically confirmed structure of General Relativity remains untouched, while its geometric meaning is interpreted within a physical field medium.

The UQSH understands itself in this framework as a possible microphysical interpretation of the geometric description of gravitation, not as its immediate replacement.

The invariance of the speed of light is thereby preserved and interpreted within the UQSH framework as an expression of the maximum reorganization rate of the field medium.

10 Falsifiability of the Approach

A field-mechanical interpretive framework must possess observable consequences that are in principle falsifiable.

The approach presented here would be particularly challenged if the following observations were clearly confirmed:

- a direct experimental detection of a consistent, universal dark matter particle species;
- additional gravitational effects that occur completely independently of baryonic structure;
- a complete explanation of the Radial Acceleration Relation within purely mass-based models without structural coupling to baryonic distributions;
- gravitational lensing effects whose spatial structure is not compatible with large-scale field geometries.

Conversely, a systematic dependence of additional gravitational effects on baryonic structure, radiation environment, large-scale field organization, or time-delayed post-organization in dynamic systems would further support a field-mechanical interpretation.

11 Discussion

The interpretation presented here does not understand itself as a complete cosmological theory, but as a conceptual approach to describing gravitational anomalies on galactic and cosmological scales.

A quantitative elaboration of this approach requires a more detailed mathematical description of the dynamics of the underlying field medium as well as a systematic comparison with astrophysical observational data.

12 Conclusion

The observed gravitational anomalies on cosmic scales are usually explained by dark matter.

The Universal Quantum Foam Hypothesis opens an alternative interpretation in which these effects can be understood as emergent properties of large-scale field organization.

The approach presented here formulates a possible conceptual framework for such an interpretation and is intended to serve as a starting point for further theoretical investigations.

A phenomenological modeling suggests that large-scale field organization can qualitatively reproduce the observed flat rotation curves.

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Note on Prior Work

An earlier monographic elaboration of related conceptual considerations by the author has been published separately. It does not constitute a formal foundation of the present work, but documents earlier developmental steps of the framework. Book on Amazon KDP: https://www.amazon.de/dp/B0GPQMX92G

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